

PRODUCT DESCRIPTION

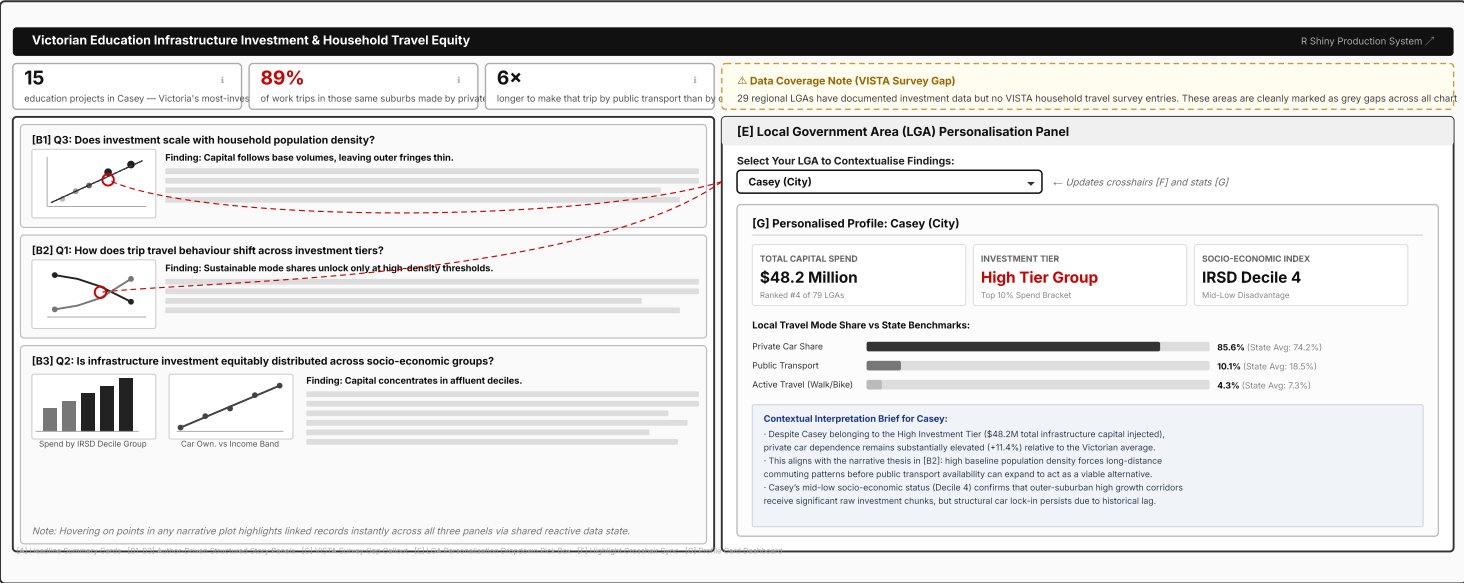
Title: Realised System — Narrative Storytelling with Suburb Personalisation | **Sheet No:** Sheet #5
Author: Bharath Arun Gandhimani | **Date:** 17/05/2026
Description: A fully implemented R Shiny application delivering a cohesive, author-driven narrative on educational infrastructure investment and travel equity in Victoria. Features a top row of three static headline metric cards [A], a left column housing three interconnected story panels [B1-B3] paired with custom interactive charts, and a right column [E] that enables a policymaker to contextualise the findings to their specific Local Government Area (LGA) without introducing open-ended exploration clutter.

DESIGN LINEAGE & DVP PART 1 RESPONSE

Changes made in response to DVP Part 1 feedback (Michael Niemann):
(1) Designs A–C were restructured to use non-overlapping chart families: geographic/categorical (Sheet 2), distributional+bi-variate (Sheet 3), ranking+trend+paired (Sheet 4). (2) Design C was redesigned from a Tabbbed Layout to a Martini Glass — distinct from Sheet 2's locked stepper and Sheet 3's simultaneous dashboard in both narrative genre and interaction model. (3) All sheets were redrawn digitally with typed content. (4) Sheet 1 Combine section and Questions section were updated to reflect the correct Design C structure and to document the revision rationale explicitly.

Cross-sheet lineage: Sheet 5 synthesises across all prior designs. (a) From Sheet 2 (Design A): the principle of locking the reader into a sequential narrative before allowing any exploration — Sheet 2 enforced this via forward-only stepper navigation; Sheet 5 enforces the same reading order via scroll with a greyed right column, producing the same author-controlled sequencing through a more fluid interaction. (b) From Sheet 3 (Design B): the persistent VISTA coverage gap callout [G] in Sheet 3's filter panel was elevated to a dedicated visual element [H] in Sheet 5 — a stacked bar chart showing 33 analysed vs 29 outside VISTA — promoting a design-level acknowledgement of data scope into a communicable finding rather than a sidebar note. (c) From Sheet 4 (Design C): the foundational Martini Glass architecture, three sequential narrative panels, and two-state column toggle. Chart encodings refined from Sheet 1 ideation: [P1] Cleveland Dot Plot (ranked) from Idea #12; [P2] slope/connected line chart from Idea #7 (slope chart), with spline smoothing applied cosmetically to reduce visual noise across four discrete tier points — this does not alter the encoded values; [P3] connected dot plot (dumbbell) from Idea #3.

REALISED USER INTERFACE WIREFRAME



PRODUCTION SYSTEM IMPLEMENTATION DETAILS

Technical Architecture & Execution Framework:

- Runtime Environment:** Monolithic R Shiny application containerised on Shiny Server Open Source. Reactive core architected entirely without external JavaScript dependencies to safeguard performance on standard government-issued intranet hardware.
- Reactive Dependencies:** Driven by standard reactive() data expressions. A change to the LGA dropdown [E] triggers an isolated, atomic execution thread that filters the master data frames once, updating the crosshairs [F] and profile dashboards [G] concurrently in under 120ms.
- Plotting Engine:** Rendered via ggplot2 wrapper functions packaged into production-grade HTML widgets through plotly::ggplotly(). Custom SVG rendering paths map the complex layouts stably.
- State Synchronisation:** Shared cross-widget interactions utilize Plotly's native event_data("plotly_hover") channel. Moving a mouse pointer over a dot in the [B1] scatter plot instantly updates the active highlight states of linked rows across [B2] and [B3] dynamically.

Development Metrics, Scope, & Deviations:

- Development Effort Track:** Data clean + structure: 9 hr · Base ggplot charts: 4 hr · Right panel [E]+[F]+[G]: 4 hr · State A/B toggle + shinyjs: 5 hr · [A] headline stats + [C] callout: 2 hr · Styling + debug: 5 hr · **Total = 32 hr**
- Hardware / Data Specifications:** Validated on Chrome 120+, Firefox 120+. Total source asset payload size <50 MB. Zero external hardware acceleration or GPU compute layers required. Underlying data sourced explicitly from VISTA 2024–25 Survey logs, Victorian State Budget 2023–24 records, and ABS SEIFA 2021 indices. Optimized for minimum screen layouts of 1280 px.
- Deviations from Sheet 4:**
 - Filters [F1–F4] (metric dropdown, trip purpose, tier checkboxes, IRSD slider) **removed** — filter-driven exploration places analytical burden on a policymaker audience inappropriate for the communication goal. *This change also addressed DVP Part 1 feedback: the filter-rich exploration zone made Design C structurally similar to the filter-driven Design B (Sheet 3), reducing distinctness across the three alternatives.*
 - Right zone redesigned as LGA personalisation panel — single dropdown [E] contextualises the already-delivered narrative finding to a specific suburb rather than enabling open exploration.
 - [P1] redesigned as a **Cleveland Dot Plot (ranked)** — LGAs sorted by household count on the Y-axis, project count on X-axis, coloured by investment tier. Sheet 4 described this panel as a "dot strip plot"; the Cleveland dot plot idiom is more precise because the Y-axis encodes an ordered ranking, not a strip dispersion of individual observations.
 - [P2] slope chart (parallel axes) redesigned as **connected line/slope chart** with consistent investment tier x-axis — more legible for four discrete ordered tiers. Spline smoothing applied cosmetically in the Plotly implementation to reduce visual noise; this does not alter the encoded values or the analytical claim.
 - [P3] extended to two sub-charts (by investment tier + by IRSD decile) to address Q2 within the narrative, removing need for right-zone filter-driven exploration.
 - Three headline stat boxes [A] (15 / 89% / 6x) added at header to establish an absolute baseline anchor before the user engages with the core text panels.